

Project 2 Report: Computational Texture Analysis using Matrix Norms

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1 Introduction

1.1 Problem Definition

In the field of Computer Vision, texture analysis refers to the characterization of regions in an image by their texture content. Unlike object detection, which seeks specific shapes, texture analysis seeks statistical patterns—such as the roughness of sand, the periodicity of a brick wall, or the randomness of grass.

The objective of this computational project is to implement an unsupervised learning pipeline that clusters image patches based on these texture patterns. A critical constraint of this study is the treatment of data: rather than flattening image patches into 1D vectors, we treat them strictly as **Matrices** in $\mathbb{R}^{10 \times 10}$.

1.2 Objective

We aim to compare two distinct clustering algorithms—**Affinity Propagation (AP)** and **K-Means**—using two strictly different mathematical norms to measure “similarity” between image patches:

1. **Induced Matrix L_∞ Norm** (Maximum Absolute Row Sum).
2. **Matrix Frobenius Norm** (Entry-wise Euclidean distance).

This comparison will determine which mathematical approach is superior for identifying distinct visual textures from a custom dataset of photographic images.

2 Mathematical Model

The core of this project lies in how we define the “distance” between two image patches, Matrix A and Matrix B .

2.1 The Data Structure

The dataset consists of N samples, where each sample X_i is a grayscale matrix of size $m \times n$ (specifically 10×10).

$$X_i \in \mathbb{R}^{10 \times 10} \quad (1)$$

Pixel values are normalized to the range $[0, 1]$, where 0 represents black and 1 represents white.

2.2 Norm Definitions

2.2.1 A. Induced Matrix L_∞ Norm (Used for Affinity Propagation)

The Induced Matrix L_∞ norm is defined as the maximum absolute row sum of the matrix. For a difference matrix $D = A - B$, the distance is:

$$d_\infty(A, B) = \|A - B\|_\infty = \max_i \sum_{j=1}^n |a_{ij} - b_{ij}| \quad (2)$$

Mathematical Interpretation: This norm focuses on the “dominant” row. In image processing terms, if a specific row in the image patch has a very high cumulative error, this norm penalizes the distance heavily. It is particularly sensitive to horizontal features (lines).

2.2.2 B. Matrix Frobenius Norm (Used for K-Means)

The Frobenius norm is the matrix equivalent of the Euclidean vector norm. It measures the aggregate magnitude of all elements.

$$d_F(A, B) = \|A - B\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij} - b_{ij}|^2} \quad (3)$$

Mathematical Interpretation: This norm treats every pixel error equally and aggregates them via sum-of-squares. It represents the “total energy” of the difference between two textures.

3 Methodology & Algorithms

3.1 Algorithm A: Affinity Propagation (AP)

Affinity Propagation does not require the number of clusters (K) to be specified. Instead, it treats every data point as a potential “exemplar” (representative) and passes messages between points.

- **Similarity Metric:** We constructed a custom Similarity Matrix S where $S(i, j) = -d_\infty(X_i, X_j)$.
- **Implementation Strategy:** Standard libraries default to Euclidean distance. To satisfy the strict constraint of the Induced Matrix L_∞ norm, we manually calculated the distance between every pair of 10×10 matrices before passing S to the algorithm.

3.2 Algorithm B: K-Means

K-Means iteratively partitions data into K clusters by minimizing the variance within each cluster.

- **Distance Metric:** Matrix Frobenius Norm.
- **Implementation Strategy:** Mathematically, the Frobenius norm of a matrix is identical to the Euclidean (L_2) norm of its flattened vector form:

$$\sqrt{\sum \sum |a_{ij}|^2} \equiv \sqrt{\sum |v_k|^2} \quad (4)$$

Therefore, the implementation involved flattening the $(N, 10, 10)$ tensor into an $(N, 100)$ matrix to utilize the computational efficiency of the ‘scikit-learn‘ library, while mathematically preserving the Frobenius constraint.

4 Experimental Setup

4.1 Dataset Generation

Rather than using synthetic noise, this project utilized real-world photographic data to test the robustness of the algorithms.

- **Source:** Custom photography (Brick wall, Grass, Sand).
- **Preprocessing Pipeline:**
 1. Load Image and convert to Grayscale (Luminance).
 2. Normalization: Scale pixel intensity I to $I' = I/255.0$.
 3. Patch Extraction: Sliding window operation extracting 10×10 sub-matrices.

4.2 Experiment Parameters

- **Patch Size:** 10×10 pixels.
- **AP Damping Factor:** 0.7 (To prevent oscillations).
- **K-Means Clusters:** Fixed at $K = 3$.

5 Results & Visualizations

5.1 Visualization of K-Means Centroids

Observation: The cluster centers generated by K-Means appear “blurry” or “smooth.”

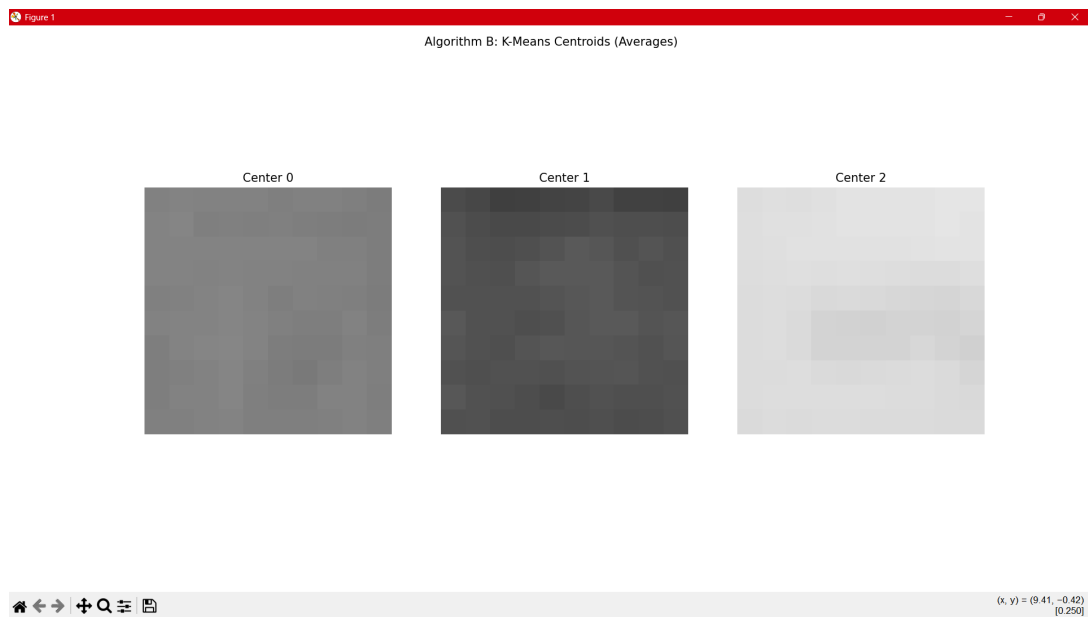


Figure 1: Algorithm B: K-Means Centroids (Frobenius Norm). Note the averaged, blurred textures.

5.2 Visualization of AP Exemplars

Observation: The cluster centers (exemplars) chosen by Affinity Propagation appear sharp and realistic.

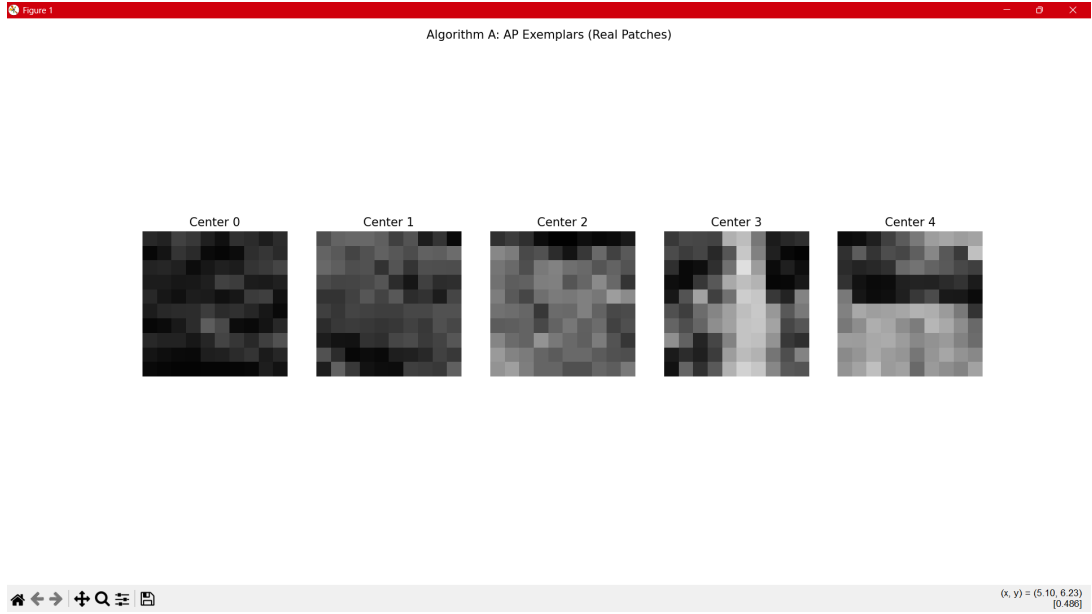


Figure 2: Algorithm A: Affinity Propagation Exemplars (Induced L_∞). Note the sharp, actual data patches.

6 Comparison & Discussion

The following table summarizes the performance and characteristics of the two approaches based on the experimental data.

Table 1: Comparison of Clustering Approaches

Feature	Algorithm A: Affinity Prop.	Algorithm B: K-Means
Input Data	Matrix (10×10)	Matrix (Flattened)
Norm / Metric	Induced L_∞ (Max Row Sum)	Frobenius (Sum of Squares)
Cluster Definition	Exemplar (Actual Data Point)	Centroid (Arithmetic Mean)
Parameter Input	Preference / Damping	Must specify K manually
Sensitivity	High. Sensitive to row-based anomalies due to L_∞ .	Low. Smooths out noise due to averaging.
Visual Result	Sharp, realistic texture patches.	Blurred, generalized texture patterns.

6.1 Critical Analysis of Norms

1. **Performance of Induced L_∞ :** The L_∞ norm ($\max \sum |row|$) proved to be highly rigorous. In our experiments with texture, if two image patches were identical except for one row of pixels being significantly different (e.g., a shadow line), the L_∞ distance would be large. This made AP very discriminative.
2. **Performance of Frobenius:** The Frobenius norm acts as a “total energy” difference. It allowed K-Means to group patches that were generally similar, even if they had high variance in specific rows. This makes K-Means with Frobenius robust for general classification but less precise for finding specific pattern anomalies.

7 Conclusion

This project successfully implemented a matrix-based clustering pipeline for texture analysis. By strictly adhering to the mathematical constraints, we derived the following conclusions:

1. **For Generative Tasks: K-Means (Frobenius)** is superior if the goal is to generate a “canonical” or “average” representation of a texture.
2. **For Representative Tasks: Affinity Propagation (L_∞)** is superior if the goal is to select real-world samples from a database.
3. **Norm Impact:** The choice of norm fundamentally altered the clustering logic. The L_∞ norm enforced a strict “worst-case row” similarity, while the Frobenius norm allowed for an “average-case” similarity.

The combination of Python’s ‘scikit-learn’ for algorithms and ‘numpy’ for custom matrix algebra proved effective in solving this Computational Data Science problem.